Practical #1

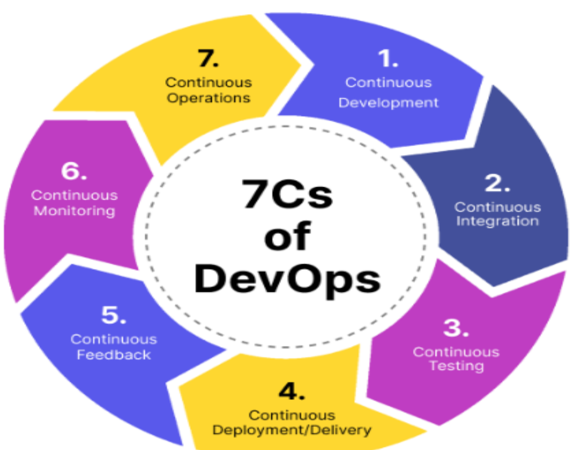
Practical Title:
To understand the DevOps culture, lifecycle, and benefits in modern software development

Theory:

1. What is DevOps?
->DevOps is a combination of **Development (Dev)** and **Operations (Ops)**.
It is a set of practices that helps developers and IT teams work together to **build, test, and release software faster and more reliably**.

2. Key Principles of DevOps
- **Collaboration:** Developers and operations teams work together
- **Automation:** Repetitive tasks like testing and deployment are automated
- **Continuous Integration (CI):** Code is regularly merged and tested
- **Continuous Delivery (CD):** Software is always ready to be released
- **Monitoring:** Continuous tracking of system performance
- **Feedback:** Quick feedback helps improve the product faster

3. DevOps Lifecycle (Draw diagram + explain each phase)



Planning: In this phase, requirements are gathered and planned. Teams decide what features to develop.
Development: Developers write code and build the application based on the plan.
Building: The code is compiled and converted into an executable application.
Testing: The application is tested automatically to find and fix bugs.
Release: The tested code is prepared for deployment.
Deployment: The application is deployed to production servers for users.

Monitoring: The system is continuously monitored for performance and issues, and feedback is collected.

Activity / Task:

Task 1:
Search and write one real-world company using DevOps
Company Name: Netflix
How DevOps is used:
Netflix uses DevOps to **continuously deliver updates** without downtime.

- Uses automation for testing and deployment
- Practices continuous integration and delivery
- Monitors systems in real-time to ensure smooth streaming
- Can release new features quickly and fix issues instantly

Task 2:
Difference between Traditional Development vs DevOps

Traditional Development	**DevOps**
Separate Dev and Ops teams	Collaboration between Dev & Ops
Slow development process	Faster development & delivery
Manual testing and deployment	Automated testing & deployment
Infrequent releases	Continuous releases
Late feedback	Continuous feedback

Advantages of DevOps:
1. Faster software delivery
2. Improved collaboration
3. Better quality products
4. Faster bug fixing
5. Increased customer satisfaction

Challenges in DevOps:
1. Cultural change in organizations
2. Requires skilled professionals
3. Tool integration can be complex